

# Kasra Mossayebi

CURRICULUM VITAE · UNIVERSITY OF TORONTO

Toronto, Ontario Canada

☎ 548-881-3896 | ✉ kasra.mossayebi@mail.utoronto.ca | 🎓 Kasra Mossayebi

## Education

### University of Toronto

Toronto, Canada

BSC IN MATHEMATICS

September 2025 -

- Pursuing mathematics specialist, 3.85 cGPA
- Took Analysis 1 (MAT157), Algebra 1 (MAT240), and Algebra 2 (MAT247)
- Member of “ $\infty$ —Categories in Algebraic Geometry” directed reading group

### Northeastern University

Boston, United States

VISITING STUDENT RESEARCHER

Summer 2026

- Official visiting student at Northeastern doing research with sponsoring faculty (INPR4100)
- Worked with members of the Institute for Artificial Intelligence in Fundamental Interactions

## Research Experience

### $A_\infty$ algebras and Entanglement

March 2026-

VISITING STUDENT RESEARCHER

- Extended entanglement DGA to an  $A_\infty$  algebra via Homotopy Transfer
- used tools from Lurie’s Higher Algebra to attempt to characterize the  $A_\infty$  algebra
- Extracted different invariants including Hochschild (co)homology and higher Massey products
- Worked at Northeastern University during the summer as a visiting student researcher

### Extending Generative effects to Entanglement

December 2024-

WITH CHRISTIAN FERKO

- Worked to extend the notion of generative effects of [Adams 2017] to detecting entangled states

### A Hodge Theory for Entanglement Cohomology

July 2024 - January 2025

WITH CHRISTIAN FERKO, EASHAN IYER, AND GREGOR SANFEY

- Built off of Mainiero’s work (arxiv: 1901.02011) to apply hodge-theoretic framework to Entanglement cohomology
- Worked on proving compatibility of support operators
- Researched extensions of entanglement cohomology and applications of poncaire polynomials
- Published in Physical Review A (PRA: 111.032422)

### Homotopical Equivalence of Categories

March 2023 - July 2023

PERSONAL PROJECT

- Furthered work done by Gurski, Johnson, and Orno (arxiv: 1508.00054) on relative categories
- Proved conditions necessary for an adjoint equivalence of relative categories to extend to an homotopical equivalence of categories
- Provided an informal talk presenting work done by Gurski et al. and my results

## Publications

[1] Christian Ferko, Eashan Iyer, Kasra Mossayebi, and Gregor Sanfey. *A Hodge Theory for Entanglement Cohomology*. Phys.Rev.A 111 (2025) 3, 032422. arXiv:2410.12529.

## Talks

### Mathematics Undergraduate Colloquium

University of Toronto

GENERATIVE EFFECTS AND QUANTUM ENTANGLEMENT

October 2025

- Presented categorical approach behind generative effects as described by [Adams 2017]
- Introduced and explained future directions for Generative effects including Segre Embedding and entanglement cohomology
- Discussed possibility of extending generative effects to descent condition for higher categories

### Physics Undergraduate Colloquium

University of Toronto

AN INTRODUCTION TO ENTANGLEMENT COHOMOLOGY

January 2026

- Introduce entanglement of the GNS complex as described in Mainiero (arxiv: 1901.02011) and Ferko et al. (2410.12529)
- Use 2-qubit example as an explicit example of why the GNS complex accurately detects entangled states